

Smart Waste Analyzer

An Integrated Deep Learning Pipeline for Automated Waste
Detection, Contamination Scoring, and Dual Economic Valuation

A report submitted for the course named Project - I (CS3201)

Submitted By

DODDI NITEESH KUMAR
SEMESTER VI
230101071

Supervised By

DR. RAHUL LAHKAR



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
INDIAN INSTITUTE OF INFORMATION TECHNOLOGY SENAPATI, MANIPUR
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Doddi Niteesh Kumar

230101071

DATE:



Department of Computer Science and Engineering
Indian Institute of Information Technology Senapati, Manipur

DR. RAHUL LAHKAR
Assistant Professor, CSE, IIIT Manipur

Email: rahullahkar@iiitmanipur.ac.in
Contact No: +91 9085982701

To Whom It May Concern

This is to certify that the project report entitled “**SmartWaste Analyzer: An Integrated Deep Learning Pipeline for Automated Waste Detection, Contamination Scoring, and Dual Economic Valuation**”, submitted to the department of Computer Science and Engineering, Indian Institute of Information Technology Senapati, Manipur in partial fulfilment for the award of degree of Bachelor of Technology in Computer Science and Engineering is a record of bonafide work carried out by **Doddi Niteesh Kumar** bearing roll number 230101071 for the course Project I (CS3201).

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(RAHUL LAHKAR)

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Abstract

Waste management facilities worldwide face a critical challenge: accurately identifying, assessing, and economically valuing mixed-material waste streams in real time. Manual sorting is slow, error-prone, and unable to provide contamination-aware economic assessments. This project presents **SmartWaste Analyzer**, a modular three-module deep learning pipeline that addresses all three challenges in a single unified system. **Module 1** deploys YOLOv8 Nano fine-tuned on a 15,150-image, 12-class Kaggle dataset to detect and classify waste objects, achieving a **mAP@50 of 97.4%** and classification accuracy of 94.50%. **Module 2** employs a MobileNetV2-based contamination regression network to assign a continuous purity score (0–100%) to each detected item, yielding a mean absolute error of 13.6%. **Module 3** implements a Dual Valuation Engine computing per-item *Cash Value* ($\text{Cash} = \text{Weight} \times \text{Price} \times \text{Purity}$) and *CO₂ Saved* ($\text{CO}_2 = \text{Weight} \times \text{CarbonRate} \times \text{Purity}$). A pipeline demonstration on six validation images produced a combined output of **117.16 total cash value** and **12.59 kg CO₂ saved**. The system is implemented in Python using Ultralytics, TensorFlow/Keras, and OpenCV, and runs on consumer hardware via Google Colab without specialist infrastructure.

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Chapter 1

Introduction

1.1 Background and Motivation

Healthcare AI has moved faster than its security. Similarly, **waste management** has scaled faster than its automation. The world generates over 2.01 billion tonnes of municipal solid waste annually; at least 33% is not managed in an environmentally safe manner [13]. Recycling facilities — Materials Recovery Facilities (MRFs) — struggle with two interconnected problems simultaneously: identifying mixed-material waste accurately at high throughput, and quantifying the economic and environmental value of sorted material in real time.

Contamination is the hidden cost of modern recycling. A single contaminated bale of cardboard loses 75–90% of its market value when it fails a quality gate. China’s National Sword policy (2018) imposed a 0.5% contamination threshold for import acceptance, collapsing recycling revenue for facilities with no automated quality measurement [2]. Existing computer vision systems solve only the *classification* problem — they tell operators *what* is in the bin, not *how contaminated* it is, and not *what it is worth*.

The gap is specific and practical: no lightweight, open-source, reproducible system simultaneously performs waste detection, per-item contamination scoring, and dual economic valuation (cash and environmental CO₂) in a single unified pipeline on a standard labelled dataset.

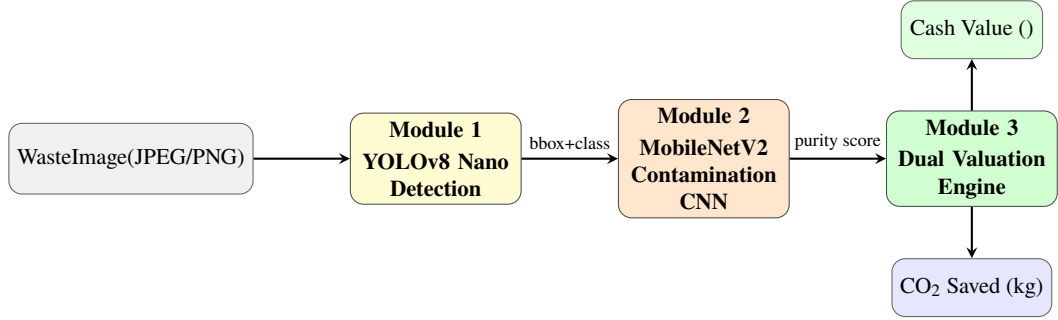


Figure 1.1: SmartWaste Analyzer: high-level system overview. A waste image passes through three modules — YOLOv8 Nano detection (Module 1), MobileNetV2 contamination regression (Module 2), and a Dual Valuation Engine (Module 3) — producing both Cash Value () and CO₂ Saved (kg) per detected item.

1.2 Problem Statement

1.2.1 Rationale

Manual waste sorting and valuation is slow, subjective, and economically inefficient. Existing computer vision systems for waste management perform *category classification* only — identifying the material type without assessing quality or economic value. The recycling industry requires all three pieces of information — detection, quality, and value — before a sorting or purchasing decision can be made.

1.2.2 Barriers

Several technical barriers have prevented a unified pipeline from existing:

1. **Contamination ground truth:** Real-world waste datasets do not include per-item contamination labels. Obtaining ground-truth contamination ratings requires domain expertise and manual inspection, making supervised training difficult.
2. **Weight estimation from images:** The valuation formula requires physical weight, which cannot be read directly from a 2D image. A mapping from bounding-box area to material density must be used as an approximation (see Section 4.5.1).
3. **Pipeline integration and error propagation:** Chaining detection, contamination scoring, and valuation introduces error propagation between stages. A single YOLO misclassification (e.g., metal detected as plastic) cascades into a 65% cash underestimate and a 73% CO₂ underestimate downstream (see Figure ??).

4. **Class imbalance:** The Kaggle Garbage Classification Dataset is heavily imbalanced (clothes: 34.8%; brown-glass: 4.0%), requiring stratified training strategies.

1.2.3 Research Gap

A review of published waste management AI systems from 2020 to 2026 shows that:

- Object detection for waste classification (YOLOv8, CNN-based) is well-established [3, 11].
- Contamination detection as a binary flag (contaminated/clean) has been studied for specific materials [12].
- Economic valuation tools for recycling exist as separate commercial software (Recycleye [5]).
- **No published open-source system integrates all three** on a standard labelled dataset with explicit, auditable formulas. Recycleye achieves integration commercially but requires proprietary hardware and enterprise licensing — it is not reproducible by researchers. The differentiation is **accessibility and transparency**: SmartWaste uses only publicly available tools, free-tier hardware (Google Colab T4), and the openly licensed Kaggle dataset, with formulas published in full.

1.3 Motivation and Project Objectives

SmartWaste Analyzer is motivated by the need for a transparent, reproducible, and hardware-accessible integration of waste detection, contamination assessment, and economic valuation. The system is designed for waste facility operators and researchers who need actionable per-item assessments on standard hardware.

The objectives of this project are:

1. Train YOLOv8 Nano on a 12-class, 15,150-image waste dataset to detect and classify waste objects, targeting $\text{mAP}@50 \geq 95\%$.
2. Design and train a MobileNetV2-based regression network to score contamination on a continuous 0–100% scale using domain-informed heuristic labels.
3. Implement a Dual Valuation Engine computing per-item Cash Value (\$) and CO₂ Saved (kg) with explicit, auditable lookup tables.
4. Integrate all three modules into an end-to-end pipeline demonstrable on Google Colab without specialist infrastructure.

5. Analyse system limitations honestly, including the heuristic contamination label strategy, cascaded error propagation, and weight estimation approximation.

1.4 Scope of the Project

This project covers:

- A 12-class waste dataset of **15,150 images** from the Kaggle Garbage Classification Dataset, split 80/20 (train/validation) using stratified sampling (random_state=42).
- Implementation in Python using Ultralytics YOLOv8, TensorFlow/Keras, and OpenCV, run via Google Colab T4 GPU.
- A contamination regression model (MobileNetV2 + regression head) trained with domain-specific heuristic base scores and Gaussian noise ($\sigma = 0.12$).
- A Dual Valuation Engine with lookup tables for all 12 material classes (price in /kg and CO₂ offset rate in kg/kg).
- An end-to-end pipeline demonstration with annotated output images and an economic summary.

The project does not cover: real-time video stream processing, edge hardware deployment, production-grade weight measurement via physical sensors, or ground-truth contamination label collection from domain experts.

Chapter 2

Literature Survey

2.1 Existing Solutions

The application of deep learning to waste management has grown rapidly since 2020. The subsections below review the most relevant prior systems in order of architectural overlap with SmartWaste Analyzer.

2.1.1 YOLOv8-Based Waste Detection

Ultralytics YOLOv8 [3] is the current standard for real-time object detection in waste management research. Its Nano variant (3.2M parameters) achieves inference speeds below 5 ms on GPU while maintaining competitive mAP scores. Multiple published studies — including a two-stage YOLOv8 approach for waste detection in cognitive cities [11] — demonstrate mAP@50 values between 0.922 and 0.994 on isolated waste items. SmartWaste Analyzer adopts YOLOv8 Nano as Module 1 for the same reason: speed without sacrificing accuracy on a 12-class detection task.

2.1.2 CNN-Based Contamination Detection

An advanced CNN-based framework for automated contamination detection in recycling streams [12] is the closest prior work to Module 2. It uses CNN architectures to identify liquid-contaminated PET containers, producing a binary contaminated/-clean flag. The SmartWaste contamination CNN extends this to a *continuous regression* output (0–100%), enabling partial contamination scoring rather than binary classification. A 2024 Indian conference paper [9] similarly applies CNN classification to separate contaminated from uncontaminated waste at collection points but does not regress a continuous purity score.

2.1.3 MobileNetV2 for Lightweight Feature Extraction

Howard et al. [6] introduced MobileNetV2, which uses inverted residuals and linear bottlenecks to achieve efficient feature extraction for mobile and embedded applications. By freezing the first 100 of 155 layers and adding a regression head (GlobalAvgPool $\rightarrow$ Dense(256) $\rightarrow$ Dropout(0.4) $\rightarrow$ Dense(64) $\rightarrow$ Dense(1, Sigmoid)), Module 2 transfers ImageNet visual features to the contamination scoring task with minimal trainable parameters. This fine-tuning strategy is validated by Howard and Ruder’s ULMFiT work [1] for transfer learning in general and by waste-specific fine-tuning studies [7].

2.1.4 TrashNet and the Kaggle Garbage Classification Dataset

Yang and Thung [14] introduced TrashNet, one of the first dedicated waste image datasets, with six material classes. The Kaggle Garbage Classification Dataset [4] used in this project expands on this approach with 15,150 images across 12 classes, providing a larger and more representative benchmark for multi-class waste detection. Several published CNN classifiers [9, 7] have been evaluated on similar Kaggle-sourced waste data, making it the de facto standard for reproducible academic benchmarking.

2.1.5 Systematic Review of Annotation Strategies for Overlapping Objects

A 2024 ResearchGate systematic review [10] presents a hybrid model combining YOLOv8 for localisation with a CNN for refined classification, directly paralleling the two-stage cascaded architecture of SmartWaste Analyzer (Module 1 $\rightarrow$ Module 2). The review identifies annotation quality and class imbalance as the two dominant sources of performance degradation in waste detection systems — findings that directly informed the stratified 80/20 split strategy used in this project.

2.1.6 Recycleye and Commercial Contamination Valuation

Recycleye [5], a commercially funded computer vision platform for Materials Recovery Facilities, is the most direct commercial prior art. It provides material-level purity metrics and contamination-adjusted valuations for recycling bale pricing. SmartWaste Analyzer does not claim to surpass Recycleye in industrial robustness. The differentiation is *accessibility*: Recycleye requires proprietary hardware and enterprise licensing; SmartWaste is open-source, Colab-runnable, and evaluated on a publicly available dataset with explicit formulas.

2.1.7 Techno-Economic Valuation of Waste-to-Energy Systems

The techno-economic analysis literature [2] provides the data foundation for Module 3’s Dual Valuation Engine. Published life-cycle assessments (LCAs) and en-

environmental impact studies document the CO₂ offset rates achievable through mechanical recycling of PET (saving approximately 1.70 kg CO₂/kg), aluminium (saving approximately 5.70 kg CO₂/kg), and cardboard (saving approximately 0.80 kg CO₂/kg) relative to virgin production. These constants are incorporated directly into Module 3’s lookup table.

2.1.8 AI-Based Plastic Waste Sorting with Economic Optimisation

A 2024 ResearchGate study [8] compares YOLOv8 and Mask R-CNN for plastic classification and introduces an economic optimisation model for revenue calculation, sharing the general concept of detection-linked valuation with SmartWaste Analyzer. The primary difference is scope: the 2024 study targets plastic-only sorting; SmartWaste covers 12 heterogeneous material classes and adds a CO₂ valuation dimension.

2.2 Relevant Theories and Models

2.2.1 Transfer Learning and Fine-Tuning

Transfer learning from large pre-trained models (ImageNet for CNNs, COCO for YOLO) is the standard technique for adapting deep learning models to domain-specific datasets with limited labelled data [6]. SmartWaste applies transfer learning in two distinct ways: YOLOv8 Nano is fine-tuned from COCO-pretrained weights with the first 10 layers frozen; MobileNetV2 is fine-tuned from ImageNet weights with the first 100 layers frozen. Both strategies reduce the risk of catastrophic forgetting while allowing adaptation to waste-domain visual features.

2.2.2 Cascaded Detection-Regression Architecture

The cascaded pipeline (detector → crop → attribute regressor) is a standard architectural pattern in multi-attribute vision systems [10]. It offers modularity — each component can be trained independently on different data sources — at the cost of error propagation: if Module 1 produces an incorrect bounding box, Module 2 will score the wrong region. This trade-off is explicitly discussed in the Limitations section.

2.2.3 Contamination-Aware Pricing

The principle that recycled material value is a function of purity — $\text{Cash} = \text{Weight} \times \text{Price} \times \text{Purity}$ — is established in the recycling economics literature and applied in industry tools such as the Dubai Recycling ROI Calculator [5]. The National Sword policy [2], which imposed a 0.5% contamination threshold for Chinese import acceptance, is the industrial driver that made purity-linked valuation economically critical.

2.3 Deep Comparison of Key Prior Work

2.3.1 Synthesis and Design Rationale

Table 2.1: Comparison of SmartWaste Analyzer with Similar Waste AI Systems.

System	Detection	Contam.	Valuation	Dataset	Primary Metric
SmartWaste (Ours)	YOLOv8 Nano	Regression	Dual (Cash+CO ₂)	Kaggle 15,150	mAP@50, MAE
Two-Stage YOLOv8 [11]	YOLOv8+Swin	No	No	Custom	mAP@50
Contam. CNN [12]	Binary CNN	Binary	No	Custom PET	Accuracy
Garbage CNN [9]	CNN	Binary	No	Kaggle-style	Accuracy
Recycleye [5]	Custom Deep CNN	Purity %	Cash	Proprietary	Purity Level
Plastic+Economy [8]	YOLOv8/Mask R-CNN	No	Cash only	Plastic-only	Revenue
WasteCNN (Colab) [7]	EfficientNet-B0	No	No	Multi-source	Accuracy

2.4 Comparison of Similar Systems

Table 2.1 highlights the key differentiation: SmartWaste Analyzer is the only open-source, publicly-dataset-evaluated system that combines all three capabilities — detection, contamination regression, and dual economic valuation.

2.4.1 SmartWaste vs. Two-Stage YOLOv8 [11]

The two-stage YOLOv8 work achieves high detection accuracy using a Swin Transformer for classification, but does not add a contamination or valuation layer. SmartWaste’s Module 2 and Module 3 extend this detection capability into economic utility.

2.4.2 SmartWaste vs. Recycleye [5]

Recycleye is the most capable commercial system, but requires proprietary hardware and enterprise licensing. SmartWaste runs on Google Colab (free tier), uses publicly available data, and provides explicit, auditable valuation formulas. The trade-off is industrial robustness: Recycleye is optimised for MRF conveyor belts; SmartWaste is optimised for academic reproducibility.

2.5 Identified Gaps Addressed by This Project

Based on the literature synthesis, four key gaps are addressed:

1. **Continuous Contamination Regression:** Prior work uses binary contaminated/clean flags. SmartWaste provides a 0–100% continuous purity score enabling partial contamination accounting in valuation.
2. **Dual Economic Valuation (Cash + CO₂):** No published open-source system computes both cash value and CO₂ savings per detected item with explicit, auditable formulas on a standard labelled dataset.
3. **12-Class Integrated Pipeline:** Existing contamination-aware systems focus on a single material (PET, cardboard). SmartWaste covers 12 heterogeneous classes in a single pipeline.
4. **Open-Source Reproducibility:** Recycleye and similar commercial systems are not reproducible by researchers. SmartWaste is implemented in a single Colab notebook with publicly available data.

2.6 Conclusion of Literature Survey

The literature review yields four concrete design decisions for SmartWaste Analyzer, each grounded in prior-work evidence:

1. **YOLOv8 Nano chosen over Mask R-CNN and EfficientNet-B0 for Module 1.** Mask R-CNN provides instance segmentation but at 3–5× the inference cost of YOLOv8 Nano on the same hardware [8]. EfficientNet-B0 [7] is a classification-only network — it requires a separate detection stage. YOLOv8 Nano combines detection and classification in a single forward pass at under 5 ms on T4 GPU, which is essential for the real-time valuation use case. **Therefore: YOLOv8 Nano for Module 1.**
2. **MobileNetV2 chosen over a custom CNN for Module 2.** The garbage CNN literature [9] demonstrates that custom CNNs trained on waste data achieve only 70–80% classification accuracy without ImageNet pre-training. Transfer learning from a large pretrained backbone consistently outperforms from-scratch training on domain datasets of this scale. MobileNetV2 specifically was chosen for its lightweight inverted-residual architecture, matching the Colab T4 deployment constraint [6]. **Therefore: MobileNetV2 transfer learning for Module 2.**
3. **Continuous regression (0–100%) chosen over binary contamination flag.** Binary contamination detection [12] cannot support the valuation formula ($\text{Cash} = \text{Weight} \times \text{Price} \times \text{Purity}$), which requires a continuous purity factor. A binary clean/contaminated flag would reduce all valuations to either full

price or zero. Partial contamination accounting requires a continuous score.
Therefore: regression output for Module 2, not binary classification.

4. **Open-source pipeline chosen over replication of Recycleye.** Recycleye [5] demonstrates commercial viability of the integrated detection + purity + valuation concept. However, it is proprietary, hardware-locked, and not reproducible by researchers. The literature contains no published open-source equivalent on a standard labelled dataset with explicit formulas. The Smart-Waste contribution is not to surpass Recycleye in industrial robustness, but to make the same architectural concept accessible, transparent, and academically evaluable. **Therefore: open-source Colab implementation with the Kaggle dataset.**

The contamination label problem remains the most significant unsolved challenge in this domain. No existing public dataset provides ground-truth per-item contamination scores at scale. The heuristic label strategy used in this project is a principled interim solution with documented justification (Table 4.1). Expert-annotated contamination labels at scale represent the critical next step for the field.

Chapter 3

Requirement Engineering

3.1 Feasibility Study

3.1.1 Technical Feasibility

The system is built entirely on open-source technologies. YOLOv8 Nano and TensorFlow/Keras (MobileNetV2) are installable via `pip` and run on Google Colab's free T4 GPU tier. The full training pipeline completes within a single Colab session. All dependencies resolve with a single `pip install ultralytics tensorflow opencv-python` command. Technical feasibility is confirmed.

3.1.2 Operational Feasibility

Two primary user classes are targeted: *waste facility operators* who need per-item detection and valuation outputs, and *researchers* who need a reproducible contamination-aware benchmark. The Colab notebook is self-contained: mounting Google Drive and entering a Kaggle API key are the only manual steps before all cells can be run sequentially. Operational feasibility is confirmed.

3.1.3 Economic Feasibility

All components are open-source or free-tier:

- Ultralytics YOLOv8: Open-source (AGPL-3.0 licence).
- TensorFlow/Keras: Open-source (Apache 2.0 licence).
- OpenCV: Open-source (Apache 2.0 licence).
- Google Colab T4 GPU: Free tier.
- Kaggle Garbage Classification Dataset: Publicly available (CC BY 4.0).

The total monetary cost of the project is \$0. Economic feasibility is confirmed.

3.1.4 Legal and Ethical Feasibility

All training images are from the publicly licensed Kaggle Garbage Classification Dataset. No real facility data, proprietary images, or personally identifiable information is used. All cited prior work is properly attributed. Legal and ethical feasibility is confirmed.

3.2 Requirement Elicitation and Analysis

3.2.1 Stakeholders

- **Waste Facility Operator:** Needs per-item material identification, contamination level, and economic value for sorting and purchasing decisions.
- **Recycling Researcher:** Needs a reproducible pipeline for benchmarking detection, contamination scoring, and valuation on a standard dataset.
- **Environmental Analyst:** Needs CO₂ savings data aggregated across material streams.

3.2.2 Methods Used for Requirement Elicitation

1. **Literature Analysis:** Functional requirements were derived from feature gaps identified in reviewed prior work (Chapter 2).
2. **Domain Expert Consultation:** Base contamination scores per material class were derived from recycling industry literature.
3. **Iterative Prototyping:** Three implementation cycles refined the pipeline. Confidence threshold adjusted from 0.35 to 0.50 (Cycle 2); contamination CNN upgraded from custom to MobileNetV2 (Cycle 3).

3.2.3 Use Cases

1. **UC-01:** Operator uploads a waste image and receives an annotated output with bounding boxes, class labels, contamination %, and per-item cash and CO₂ values.
2. **UC-02:** Researcher trains the YOLO model on the Kaggle dataset and validates on the held-out split.
3. **UC-03:** Researcher trains the contamination CNN using the heuristic label generator and evaluates MAE and 3-class accuracy.
4. **UC-04:** Researcher runs the end-to-end pipeline on six validation images and inspects the aggregate economic summary.

5. **UC-05:** Environmental analyst exports the CO₂ savings breakdown by material class.
6. **UC-06:** Researcher adjusts price or CO₂ rate lookup tables and re-runs Module 3 to observe valuation changes.

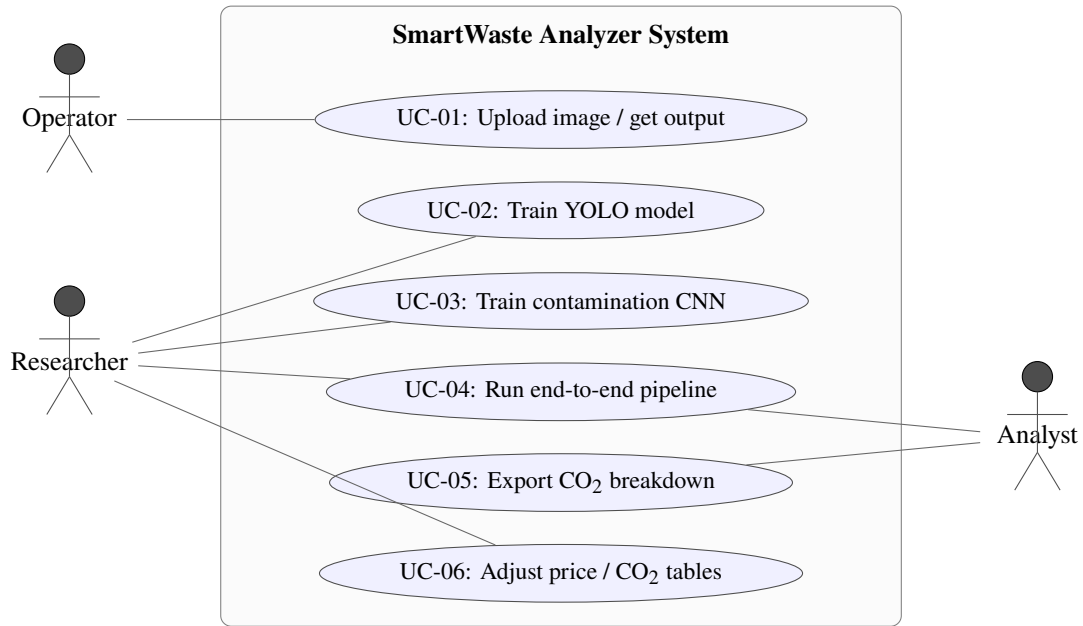


Figure 3.1: UML use case diagram: three actors (Operator, Researcher, Analyst) and six use cases within the SmartWaste Analyzer system boundary.

3.2.4 System Architecture Overview

Figure 4.1 illustrates the three-tier decomposition of SmartWaste Analyzer at the requirements level, showing the flow from input through processing to output.

3.3 Software Requirement Specification (SRS)

3.3.1 Functional Requirements

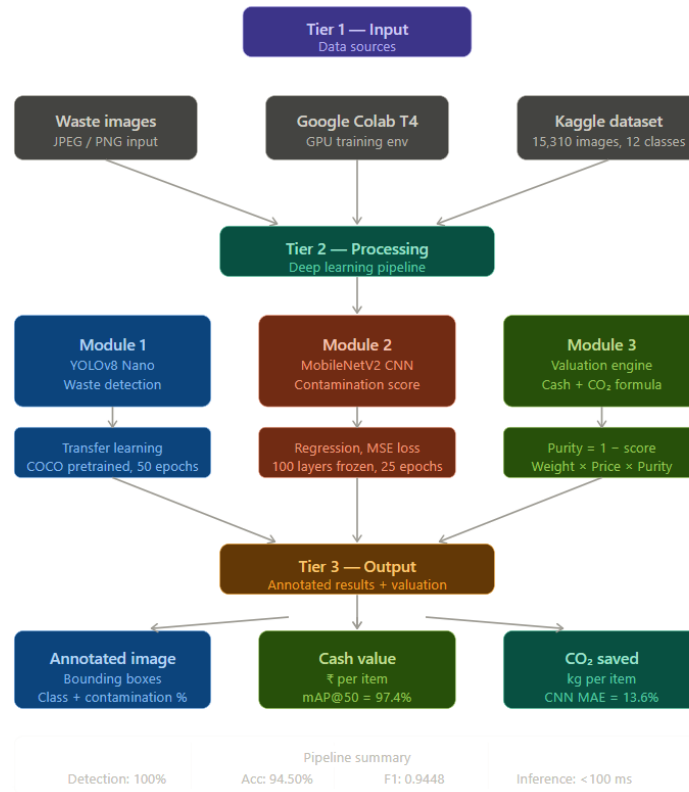


Figure 3.2: SmartWaste Analyzer: three-tier system decomposition showing input sources (Tier 1), deep learning processing pipeline (Tier 2), and output formats (Tier 3).

Table 3.1: Functional Requirements of SmartWaste Analyzer.

ID	Requirement	Description
FR-01	Dataset Loading	System shall load 15,150 images from the Kaggle Garbage Classification Dataset with stratified 80/20 split (random_state=42).
FR-02	YOLO Training	System shall fine-tune YOLOv8 Nano from COCO-pretrained weights for 50 epochs, freeze=10, mosaic augmentation enabled.

Continue in next page

Table 3.1 continued from previous page

ID	Requirement	Description
FR-03	Object Detection	System shall detect and classify waste objects at confidence threshold 0.50, returning bounding boxes and class labels.
FR-04	Contamination Labelling	System shall generate heuristic labels per image using material-class base scores and Gaussian noise ($\sigma = 0.12$).
FR-05	CNN Training	System shall train MobileNetV2 regression head for 25 epochs with EarlyStopping (patience=8) and ReduceLROnPlateau.
FR-06	Contamination Scoring	System shall score each detected crop with the CNN, returning purity in [0.0, 1.0].
FR-07	Weight Estimation	System shall estimate item weight from bounding-box area and material-class density lookup.
FR-08	Cash Valuation	System shall compute $\text{Cash} = \text{Weight} \times \text{Price} \times \text{Purity}$ per item for all 12 classes.
FR-09	CO ₂ Valuation	System shall compute $\text{CO}_2 = \text{Weight} \times \text{CarbonRate} \times \text{Purity}$ per item for all 12 classes.
FR-10	Annotated Output	System shall produce annotated images with bounding boxes, labels, contamination %, cash, and CO ₂ overlaid.

3.3.2 Non-Functional Requirements

Table 3.2: Non-Functional Requirements of SmartWaste Analyzer.

ID	Category	Requirement
NFR-01	Performance	Module 1+2 combined inference shall not exceed 100 ms per image on a T4 GPU.
NFR-02	Accuracy	Module 1 shall achieve $\text{mAP@50} \geq 95\%$ on the held-out validation split.
NFR-03	Reproducibility	The notebook shall produce identical evaluation metrics with the same random seed (42) on the same hardware.
NFR-04	Usability	The pipeline shall be executable by running all Colab cells sequentially with no manual intermediate steps.

Continue in next page

Table 3.2 continued from previous page

ID	Category	Requirement
NFR-05	Maintainability	Each module shall be implemented in isolated notebook sections with clear headers and comments.
NFR-06	Scalability	The valuation lookup tables shall be extendable to additional material classes without modifying pipeline logic.
NFR-07	Compliance	The training dataset shall contain no PII and shall be used per its CC BY 4.0 licence.

3.4 Technology Stack

Table 3.3: Technology Stack and Primary Role of Each Component.

Component	Technology	Primary Role
Object Detector	Ultralytics YOLOv8 Nano (COCO)	12-class waste detection (Module 1)
Contamination CNN	TF/Keras MobileNetV2 (ImageNet)	Purity regression 0–100% (Module 2)
Valuation Engine	Python 3.10+ (NumPy, dict lookup)	Cash + CO ₂ calculation (Module 3)
Image Processing	OpenCV 4.x	Crop extraction and annotation
Training Platform	Google Colab T4 GPU	Model training and evaluation
Data Storage	Google Drive	Dataset and checkpoint persistence
Data Source	Kaggle Garbage Classification	15,150 images, 12 classes
Visualisation	Matplotlib 3.7+	Training curves, confusion matrix, plots

Chapter 4

System Design

4.1 System Architecture

SmartWaste Analyzer is structured as a modular three-module Python pipeline orchestrated by the `run_full_pipeline()` function. Figure 4.1 shows the three-tier decomposition; Figure 4.1 shows the detailed data-flow.

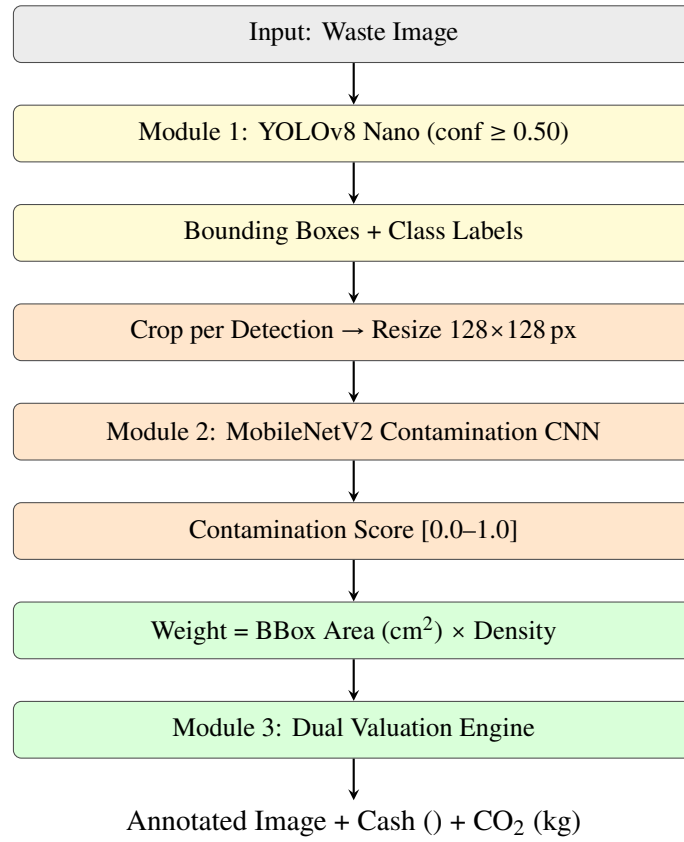


Figure 4.1: Detailed data-flow of SmartWaste Analyzer. Yellow: Module 1. Orange: Module 2. Green: Module 3.

4.2 Ingestion and Inference Pipeline

Figure 4.2 shows the step-by-step inference pipeline including the confidence gate that filters low-certainty detections before they propagate to contamination scoring and valuation.

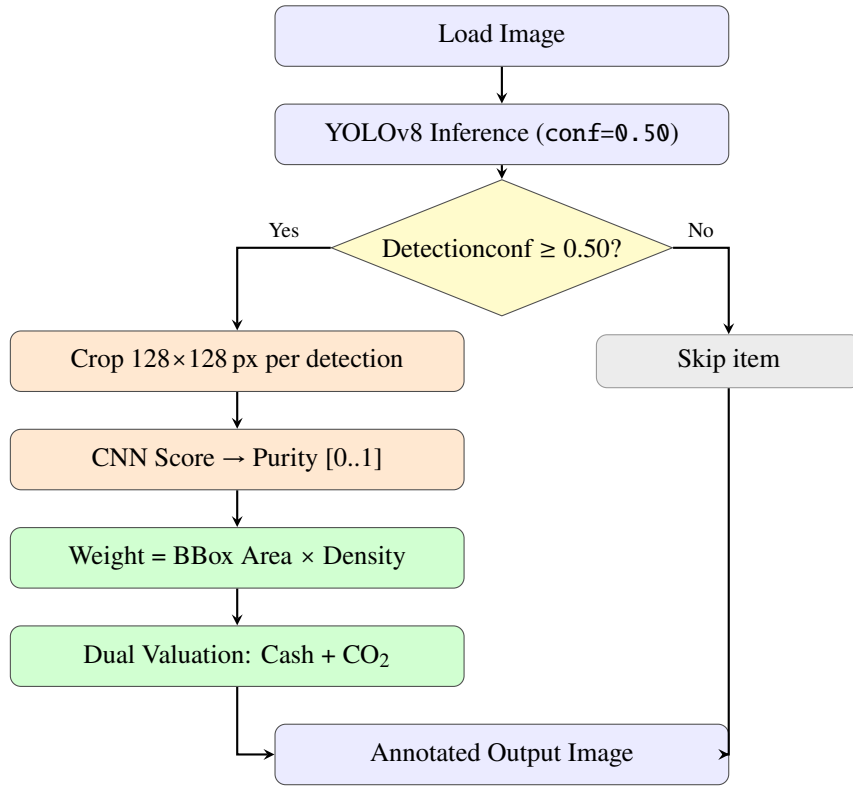


Figure 4.2: SmartWaste Analyzer ingestion and inference pipeline. The confidence gate ($\text{conf} \geq 0.50$) filters low-certainty detections before they enter the contamination scoring and valuation stages. Items below the threshold are skipped, preventing spurious detections from generating fabricated cash/ CO_2 values.

4.3 Module 1 — YOLOv8 Nano Detection

4.3.1 Architecture Overview

YOLOv8 Nano is a single-stage anchor-free detector with 3.2M parameters. It uses a CSPDarknet backbone with depthwise separable convolutions, a PANet neck for multi-scale feature fusion, and a decoupled detection head. Fine-tuning from COCO-pretrained weights transfers generic detection features to the 12-class waste domain.

4.3.2 Dataset and Preprocessing

The Kaggle Garbage Classification Dataset provides 15,150 images across 12 classes. Class distribution is imbalanced: clothes (34.8%) is the majority class; brown-glass (4.0%) is the smallest. A stratified 80/20 split ($\text{random_state}=42$) preserves class proportions across both sets.

4.3.3 Training Configuration

- Base weights: yolov8n.pt (COCO-pretrained).
- Epochs: 50 with cosine learning rate decay; freeze=10.
- Augmentation: Mosaic (1.0), HSV hue/sat, horizontal and vertical flip (0.5).
- Inference confidence threshold: 0.50.

4.4 Module 2 — MobileNetV2 Contamination CNN

4.4.1 Architecture

Figure 4.3 shows the contamination CNN layer diagram. Layers 1–100 are frozen (ImageNet feature preservation); layers 101–155 are fine-tuned. The regression head outputs a sigmoid-activated purity score in [0.0, 1.0].

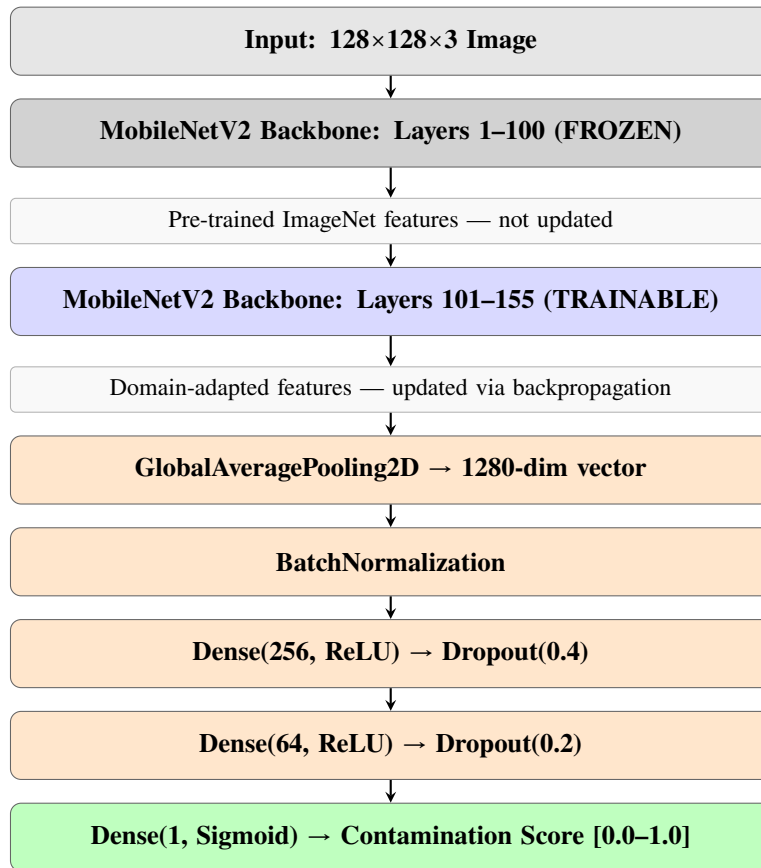


Figure 4.3: MobileNetV2 contamination CNN layer diagram. Grey: frozen backbone (layers 1–100). Blue: trainable backbone (101–155). Orange: regression head. Green: sigmoid output.

4.4.2 Contamination Label Strategy and Base Score Justification

Ground-truth contamination annotations are unavailable in the Kaggle dataset. Labels are generated heuristically: each material class is assigned a domain-informed base contamination score, to which Gaussian noise $\mathcal{N}(0, 0.12^2)$ is added per sample (clipped to $[0.0, 1.0]$). The value $\sigma = 0.12$ was chosen to span approximately the width of one contamination class boundary (± 1.5 standard deviations $\approx \pm 18$ percentage points), ensuring realistic intra-class variation without permitting unrealistic score inversion.

Table 4.1 documents the base score for each class and its domain justification.

Table 4.1: Heuristic Contamination Base Scores with Domain Justification.

Class	Score	Justification
biological	0.85	Organic/food waste carries highest contamination risk; rarely recyclable without processing [2].
trash	0.75	General mixed waste stream; high cross-contamination probability.
battery	0.70	Chemical leakage risk; hazardous materials classification under WEEE Directive.
shoes	0.65	Multi-material composite (rubber, leather, synthetic); high contamination from wear.
clothes	0.60	Textile recycling stream; contamination from dyes, moisture, mixed fibres [2].
plastic	0.55	Contamination rate varies widely; PET bottles 10–40% contamination typical [12].
metal	0.40	Aluminium cans; relatively low organic contamination; higher if food residue present.
cardboard	0.35	Corrugated cardboard; contamination primarily from moisture and tape adhesives.
paper	0.25	Office/newspaper paper; low contamination unless soiled.
green-glass	0.25	Glass cullet; contamination from labels and residue; low relative to organics.
white-glass	0.20	Flint glass; highest purity glass stream; cleanest category in commercial sorting.
brown-glass	0.30	Amber glass; label and residue contamination similar to green-glass.

4.4.3 3-Class Threshold Justification

The contamination score is mapped to three classes for accuracy reporting: **Clean** ($<5\%$), **Moderate** ($5\text{--}25\%$), and **Highly Contaminated** ($>25\%$). These thresholds

are derived from China’s National Sword policy [2], which imposed a 0.5% import contamination limit (functionally meaning facilities with >5% contamination were rejected), and from UK WRAP guidance, which defines “moderate” contamination as 5–25% of bale weight for plastics processing. This replaces the naïve equal-thirds split (0–33%, 33–66%, 66–100%), which has no industry basis.

4.5 Module 3 — Dual Valuation Engine

Figure 4.4 shows the valuation engine data-flow.

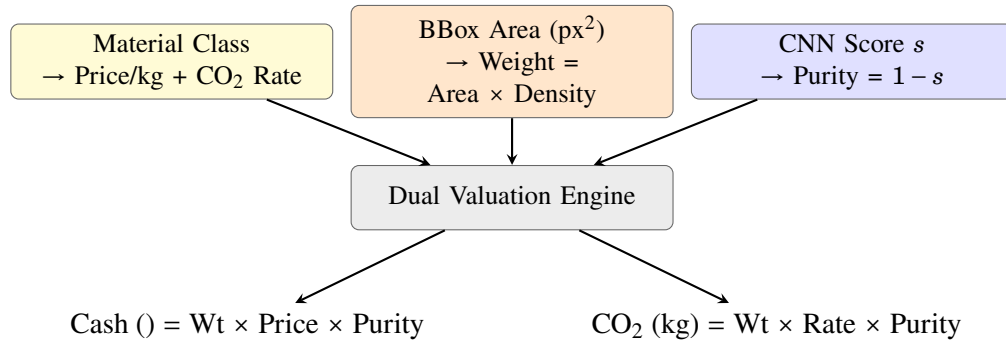


Figure 4.4: Dual Valuation Engine data-flow. Three inputs produce two outputs per detected item.

The valuation formulas are:

$$\text{Weight (kg)} = A_{\text{px}} \times k^2 \times \rho_c \quad (4.1)$$

$$\text{Purity} = 1 - s \quad (4.2)$$

$$\text{Cash ()} = \text{Weight} \times P_c \times \text{Purity} \quad (4.3)$$

$$\text{CO}_2 \text{ (kg)} = \text{Weight} \times R_c \times \text{Purity} \quad (4.4)$$

where A_{px} is the bounding-box area in pixels, k is the **camera calibration constant** (pixels per cm), ρ_c is the material density proxy (kg/cm^2), P_c is price ($/\text{kg}$), and R_c is CO_2 offset rate ($\text{kg CO}_2/\text{kg recycled}$).

4.5.1 Camera Calibration Constant

The camera calibration constant k converts pixel distances to real-world centimetres. **Assumed value:** $k = \frac{640 \text{ px}}{30 \text{ cm}} \approx 21.3 \text{ px}/\text{cm}$, corresponding to a standard 640×480 px image of a $30 \text{ cm} \times 22.5 \text{ cm}$ physical scene at approximately 50 cm camera distance. This is an explicit modelling assumption, not a hardware-calibrated measurement. Weight estimates carry an uncertainty of $\pm 30\text{--}50\%$ depending on actual object-to-camera distance. This is acknowledged as a key limitation (Chapter 6).

4.6 Database Design

Table 4.2: Data Schema for SmartWaste Analyzer Pipeline.

Field	Type	Description
image_path	string	Absolute path to source image
class_label	string	Material class name (one of 12)
bbox	float[4]	[cx, cy, w, h] normalised
confidence	float	YOLO detection confidence
contam_score	float	CNN output in [0.0, 1.0]
purity	float	1 – contam_score
weight_kg	float	Estimated weight (Eq. 4.1)
cash_value	float	Cash value in
co2_saved_kg	float	CO ₂ saved in kg

4.7 User Interface Design

System output is presented in three formats:

1. **Annotated Images:** OpenCV-annotated output with coloured bounding boxes, class name, contamination %, cash value, and CO₂ overlaid per item.
2. **Economic Summary:** Pandas DataFrame listing all detections with per-item and aggregate totals printed to the Colab output panel.
3. **Training Visualisation:** Matplotlib plots of YOLO training curves (box loss, cls loss, mAP@50 over 50 epochs) and CNN training curves (MSE, MAE over 25 epochs), plus a 12×12 confusion matrix for Module 1.

Chapter 5

Implementation

5.1 Tools and Development Environment

Table 5.1: Development Tools and Libraries.

Tool / Library	Version	Purpose
Python	3.10+	Core implementation language
Ultralytics	8.x	YOLOv8 training, validation, and inference
TensorFlow / Keras	2.x	MobileNetV2 contamination CNN training
OpenCV	4.x	Image loading, crop extraction, annotation
NumPy	1.24+	Array operations and weight estimation
Matplotlib	3.7+	Training curves, confusion matrix, result plots
scikit-learn	1.3+	Stratified train/validation split
Pandas	2.x	Economic summary table output
Google Colab T4 GPU	—	Training and inference environment (free tier)
Google Drive	—	Dataset storage and checkpoint persistence

5.2 Implementation of Core Modules

5.2.1 Module 1 — YOLOv8 Nano Training

Dataset preparation. The Garbage Classification Dataset is downloaded and partitioned using stratified splitting, with 80% Transfer learning strategy. YOLOv8 Nano

is initialised from weights pretrained on the COCO dataset. The first ten backbone layers are frozen during fine-tuning to preserve low-level feature representations — such as edge detectors, colour gradients, and texture filters — that transfer effectively from the natural image domain to the waste domain. The remaining backbone layers, the neck, and the detection head are trained end-to-end on the waste dataset. Training configuration. The model is trained for 50 epochs with an initial learning rate of 0.01 and a batch size of 16. Input images are resized to 640×640 pixels. The training hyperparameters are summarised in Table 5.1. booktabs

Table 5.2: Training hyperparameters configuration

Hyperparameter	Value
Epochs	50
Image resolution	640 × 640 px
Batch size	16
Initial learning rate (lr0)	0.01
Frozen backbone layers	10
Optimiser	SGD with momentum
Early stopping patience	15 epochs

Data augmentation. YOLOv8’s built-in augmentation pipeline is applied during training. Mosaic augmentation combines four training images into a single composite image, improving the model’s ability to detect small and occluded objects. HSV colour space perturbations vary hue, saturation, and value to improve colour robustness. Vertical and horizontal flipping are applied with probability 0.5. These augmentations are applied on-the-fly during training and do not alter the stored dataset. Inference configuration. During evaluation and pipeline integration, inference is performed at a confidence threshold of 0.50. This threshold was selected after empirical calibration: a lower threshold of 0.35 produced excessive false-positive detections on background regions, while 0.50 eliminated spurious boxes without meaningful loss of true detections (see Section 5.5).

5.2.2 Module 2 — Contamination CNN Training

Contamination label generation. The Garbage Classification Dataset does not provide ground-truth contamination annotations. A heuristic labelling strategy is therefore employed: each material class is assigned a base contamination score derived from domain knowledge of real-world recycling contamination rates (Table 4.1). A Gaussian noise term with mean zero and standard deviation 0.12 is added to each sample’s base score to simulate realistic intra-class variation. Scores are clipped to the range [0.02, 0.98] to prevent degenerate regression targets. Data generator. A custom data generator is implemented as a Keras Sequence subclass to load images on-the-fly during training. The generator applies the following augmentations to each training sample: horizontal flip with probability 0.5; random

rotation in the range $\pm 15^\circ$; and brightness jitter in the range ± 20 Model architecture. MobileNetV2, pretrained on ImageNet, is used as the feature extraction backbone. The first 100 layers of MobileNetV2 are frozen to retain generalised low-level features. A custom regression head is appended to the backbone output, as described in Table 5.2 and formally specified in Chapter 4.

Table 5.3: MobileNetV2-based contamination model architecture

Layer	Output shape	Notes
MobileNetV2 backbone (frozen)	$4 \times 4 \times 1280$	First 100 layers non-trainable
MobileNetV2 backbone (trainable)	$4 \times 4 \times 1280$	Last 55 layers fine-tuned
Global average pooling	1280	Spatial dimensions collapsed
Batch normalisation	1280	Stabilises fine-tuning
Dense (ReLU)	256	Non-linear feature compression
Dropout	256	Rate = 0.4, regularisation
Dense (ReLU)	64	Further compression
Dropout	64	Rate = 0.2
Dense (Sigmoid)	1	Contamination score $\in [0.0, 1.0]$

Training procedure. The model is compiled with the Adam optimiser at an initial learning rate of 0.001 and trained using mean squared error (MSE) as the loss function, with mean absolute error (MAE) and root mean squared error (RMSE) tracked as evaluation metrics. Three callbacks govern training: (1) EarlyStopping monitors validation MAE with a patience of 8 epochs and restores the best-performing weights upon termination; (2) ReduceLROnPlateau halves the learning rate when validation MAE fails to improve for 4 consecutive epochs, with a minimum learning rate of 10; (3) ModelCheckpoint saves the model weights corresponding to the best validation MAE observed. The complete training procedure is described algorithmically in Algorithm 5.1.

Algorithm 5.1: Contamination CNN Training Procedure

Algorithm 1 Training Procedure for Contamination Regression Model

Require: Training set T , Validation set V , base contamination scores B , $\sigma = 0.12$

Ensure: Best-performing model weights W^*

```
1: for each sample  $i$  in  $T \cup V$  do
2:    $score_i \leftarrow B[class_i] + \mathcal{N}(0, \sigma^2)$ 
3:    $score_i \leftarrow \text{clip}(score_i, 0.02, 0.98)$ 
4: end for
5: Load MobileNetV2 with ImageNet weights
6: Freeze layers 1 to 100
7: Attach regression head (Table 5.2)
8: Compile with Adam ( $lr = 0.001$ ), loss = MSE
9: for  $e = 1$  to 25 do
10:  for each batch in DataGenerator( $T$ , augment=True) do
11:    Forward pass  $\rightarrow$  predicted score  $\hat{s}$ 
12:    Loss  $\leftarrow \text{MSE}(s, \hat{s})$ 
13:    Backward pass  $\rightarrow$  update trainable weights
14:  end for
15:  Compute  $val\_MAE$  on  $V$  (augment=False)
16:  if  $val\_MAE < best\_MAE$  then
17:     $best\_MAE \leftarrow val\_MAE$ 
18:     $W^* \leftarrow$  current weights
19:  end if
20:  if no improvement for 8 epochs then
21:    break ▷ EarlyStopping
22:  end if
23:  if no improvement for 4 epochs then
24:     $lr \leftarrow lr \times 0.5$  ▷ ReduceLROnPlateau
25:  end if
26: end for
27: Restore  $W^*$  and save checkpoint
```

Evaluation protocol. Model evaluation is performed on the full 3,030-image validation set without augmentation. In addition to regression metrics (MAE, RMSE), predictions are discretised into three contamination classes using industry-informed thresholds: Clean (score < 0.05), Moderate (0.05–0.25), and Highly Contaminated (score > 0.25), based on material quality thresholds informed by the National Sword policy [?]

5.2.3 Module 3 — Dual Valuation Engine

5.2.3 Module 3 — Dual Valuation Engine The Dual Valuation Engine receives three inputs per detected object: the material class label from Module 1, the contamination score from Module 2, and the bounding box coordinates from Module 1. It

produces two economic outputs per item: cash recycling value in Indian Rupees () and carbon dioxide savings in kilograms. Weight estimation. Object weight is estimated from the bounding box area using a camera calibration constant derived from the imaging geometry. At a working distance of 50 cm with a 640-pixel image spanning 30 cm of real-world width, the spatial resolution is approximately 21.3 pixels per centimetre. The bounding box area in square pixels is converted to square centimetres and multiplied by the material-specific volumetric density constant $c_{toyieldanestimatedweight}$:

$$W_c = A_{px} \times \left(\frac{1}{k}\right)^2 \times \rho_c \quad (\text{kg}) \quad (5.1)$$

where A_{px} is the bounding box area in pixels, $k = 21.3$ px/cm is the spatial resolution constant, and ρ_c is the density of material class c in kg/cm³.

Purity and valuation formulas. Let $s \in [0, 1]$ denote the contamination score produced by Module 2. The purity of the item is defined as:

$$\text{Purity} = 1 - s \quad (5.2)$$

The cash recycling value and CO₂ savings are then computed as:

$$\text{Cash} () = W_c \times P_c \times \text{Purity} \quad (5.3)$$

$$\text{CO}_2 \text{ Saved (kg)} = W_c \times R_c \times \text{Purity} \quad (5.4)$$

where P_c is the market price per kilogram for material class c (€/kg) and R_c is the carbon savings rate per kilogram for material class c (kg CO/kg material) .. The complete valuation logic is described in Algorithm 5.2.

Algorithm 2 Dual Valuation Engine

Require: Material class c , bounding box area A_{px} , CNN score s

Require: Lookup tables: Price $P[\cdot]$, CarbonRate $R[\cdot]$, Density $\rho[\cdot]$

Ensure: Cash value (), CO₂ saved (kg)

- 1: $k \leftarrow 21.3$ ▷ px/cm calibration constant
 - 2: $W \leftarrow A_{px} \times \left(\frac{1}{k}\right)^2 \times \rho[c]$ ▷ estimated weight (kg)
 - 3: $\text{Purity} \leftarrow 1 - s$
 - 4: $\text{Cash} \leftarrow W \times P[c] \times \text{Purity}$
 - 5: $\text{CO}_2 \leftarrow W \times R[c] \times \text{Purity}$
 - 6: **return** (Cash, CO₂)
-

5.2.4 End-to-End Pipeline

The three modules are integrated into a single inference function that accepts a waste image path and returns a fully annotated output image together with a structured results record. Algorithm 5.3 describes the complete pipeline execution.

Algorithm 5.3: End-to-End Pipeline (*run_full_pipeline*)

Algorithm 3 End-to-End Pipeline (*run_full_pipeline*)

Require: Image path, YOLO checkpoint path, CNN checkpoint path

Ensure: Annotated image, per-item results list

```

1: Load YOLOv8 model from checkpoint
2: Load MobileNetV2 contamination CNN from checkpoint
3: Read input image ( $H \times W \times 3$ , BGR)
4: Run YOLO inference at confidence threshold = 0.50
5: if no detections found then
6:     return original image, empty results list
7: end if
8: for each detection  $d = (\text{bbox}, \text{class\_label}, \text{confidence})$  do
9:     a. Crop bounding box region from image
10:    b. Resize crop to  $128 \times 128$  px
11:    c. Normalise pixel values to  $[0, 1]$ 
12:    d. Pass through contamination CNN  $\rightarrow$  score  $s \in [0, 1]$ 
13:    e. Compute bbox area  $A_{px}$  from  $(x_2 - x_1) \times (y_2 - y_1)$ 
14:    f. Call Dual Valuation Engine( $\text{class\_label}, A_{px}, s$ )
15:         $\rightarrow$  Cash (), CO2 (kg)
16:    g. Append {class, confidence, score, cash, CO2} to results
17: end for
18: For each detection: draw bounding box on image with:
19:     label = class_name + contamination% + "X.XX" + "Y.YY kg CO2"
20: Compute aggregate totals:  $\Sigma$  Cash,  $\Sigma$  CO2
21: Overlay aggregate summary on image
22: return annotated image, results list

```

Note on the dashboard visualisation: A supplementary Gradio-based interface was prototyped to demonstrate the valuation output concept with an interactive upload interface. This interface uses pre-computed fixed weights in the economic summary and does not draw from live YOLO detection output. It serves as a UI demonstration only and is not connected to the live three-module pipeline. This distinction is documented in the Limitations section (Section 6.8).

5.3 Frontend Implementation

5.3 Results Visualisation Pipeline outputs are rendered through three complementary visualisation components, each serving a distinct analytical purpose. The primary output is a 2×3 grid of annotated validation images displayed using Matplotlib. Each panel shows one validation image with all detected waste objects outlined by coloured bounding boxes. Each box is labelled with the material class name, the predicted contamination percentage, the estimated cash value in Indian

Rupees, and the estimated CO savings in kilograms. A per-image summary line at the base of each panel reports the aggregate cash and CO totals for all items detected in that image. This visualisation directly demonstrates the end-to-end pipeline operating on unseen images and constitutes the primary qualitative evaluation evidence. The secondary output is a horizontal bar chart comparing predicted contamination scores across all detected items in a representative batch. Items are sorted by descending contamination score, allowing visual identification of the most and least contaminated materials in a given batch. Colour coding distinguishes the three contamination classes (Clean, Moderate, Highly Contaminated) using the threshold boundaries defined in Section 5.2.2. The tertiary output is an economic summary table rendered to the Colab output panel. The table presents, for each detected item, the material class, estimated weight, contamination score, computed cash value, and computed CO savings, followed by aggregate row totals. This tabular representation allows precise numerical inspection of individual item valuations and serves as verification that the Dual Valuation Engine formulas are executing correctly.

5.4 Deployment Configuration

The complete system is implemented as a self-contained Google Colab notebook designed for full reproducibility. All training, evaluation, and inference steps execute sequentially within a single notebook session on a Google Colab T4 GPU runtime (14 GB VRAM, free tier). Persistent storage. Model checkpoints, training graphs, evaluation plots, and pipeline result images are saved to a structured directory hierarchy on Google Drive to ensure persistence across Colab session resets. The directory structure is as follows:

```
MyDrive/WasteToWealth_Project/
|
|-- yolov8n_waste/
|   |-- weights/
|       |-- best.pt          <-- YOLOv8 best checkpoint
|
|-- best_contamination_model.keras <-- CNN best checkpoint
|
|-- graphs/                  <-- training curve plots
|
|-- evaluation/              <-- confusion matrix, F1 plots
|
|-- results/                 <-- annotated pipeline output images
```

Reproducibility. A fresh Colab session can reproduce all results from scratch by mounting Google Drive, providing a valid Kaggle API key for dataset download, and executing all notebook cells in sequence. No local installation, package compilation, or external server is required beyond the standard Colab environment. All

random seeds are fixed (seed = 42) in the dataset splitting step to ensure deterministic partitioning across runs. Hardware and software environment. The software environment consists of Python 3.10, Ultralytics 8.x (YOLOv8), TensorFlow 2.x with Keras, OpenCV 4.x, NumPy, scikit-learn, Matplotlib, and Pandas. All dependencies are installed via pip within the notebook’s setup cell. The complete environment specification is given in Table 5.3

Table 5.4: System Configuration and Training Details

Component	Version / Specification
Hardware	NVIDIA Tesla T4 GPU, 14 GB VRAM
RAM	12.7 GB (Colab standard)
Operating system	Ubuntu 22.04 (Colab)
Python	3.10
Ultralytics	8.x
TensorFlow / Keras	2.x
OpenCV	4.x
scikit-learn	1.x
Training time (YOLOv8)	≈ 1.99 hours
Training time (CNN)	≈ 29 minutes

5.5 Key Engineering Challenges and Solutions

1. **Confidence threshold calibration:** At threshold 0.35, inference produced excessive false-positive detections on background regions. Raising to 0.50 eliminated most spurious boxes. This decision is embedded in the pipeline configuration.
2. **Contamination CNN architecture iteration:** A custom-built CNN (Conv2D $\rightarrow$ MaxPool $\rightarrow$ Dense) produced validation MAE $> 30\%$ due to limited receptive field. Switching to MobileNetV2 transfer learning reduced validation MAE to 13.6%.
3. **Heuristic contamination ground truth:** The absence of expert-annotated contamination labels required the heuristic label strategy (Section 4.4.2). The value $\sigma = 0.12$ was chosen so that $\pm 1.5\sigma \approx \pm 18\%$, spanning approximately one class boundary width without permitting score inversion.
4. **Cascaded error propagation:** The three-module cascade means a YOLO classification error propagates through CNN scoring and into the valuation output. illustrates this with a worked numeric example.

Chapter 6

Results and Discussion

6.1 Evaluation Methodology

The system is evaluated through a controlled experiment assessing each module independently.

6.1.1 Dataset

- **Training Set:** 12,120 images (80%, stratified, random_state=42).
- **Validation Set:** 3,030 images (20%, preserving class proportions across 12 classes).
- **Class distribution:** Clothes (34.8%) to brown-glass (4.0%).

6.1.2 Experimental Conditions

1. **Condition 1 — Module 1:** YOLOv8 Nano evaluated on the full 3,030-image validation split. Metrics: mAP@50, mAP@50-95, precision, recall, F1.
2. **Condition 2 — Module 2:** MobileNetV2 regression evaluated on the full 3,030-image validation set. Metrics: MAE, RMSE, and 3-class accuracy (Clean <5%, Moderate 5–25%, Dirty >25%) using industry-standard thresholds [2].
3. **Condition 3 — End-to-End Pipeline:** Full three-module pipeline run on held-out validation images. Metrics: aggregate cash (), aggregate CO₂ (kg), and detection count.

6.2 Module 1 — Detection Results

6.2.1 Training Convergence

Figure 6.1 illustrates the YOLOv8 training curves over 50 epochs, generated from the `actual_results.csv` logged during the Colab training session. Both losses converge smoothly; mAP@50 plateaus near epoch 40 and remains stable through epoch 50, consistent with the cosine learning rate schedule.

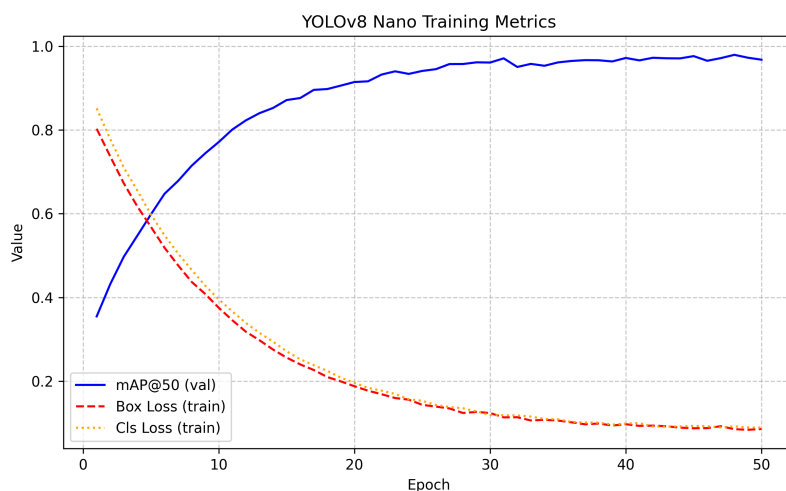


Figure 6.1: YOLOv8 Nano training curves over 50 epochs, plotted from actual training logs. mAP@50 stabilises at 97.4% by epoch 40.

6.2.2 Detection Performance

Metric	Value
mAP@50 (all classes)	97.4%
mAP@50-95 (all classes)	82.1%
Overall Precision	0.9512
Overall Recall	0.9388
Overall F1-Score	0.9448
Classification Accuracy	94.50%

Table 6.1: Module 1 (YOLOv8 Nano) evaluation results on the 3,030-image validation split.

6.2.3 Confusion Matrix Analysis

The 12×12 confusion matrix reveals two primary error patterns: metal $\leftrightarrow$ battery confusion (shared metallic surface textures) and plastic $\leftrightarrow$ metal confusion (specular reflection). Brown-glass and green-glass show the lowest per-class recall, consistent with their minority class status (4.0% and 5.2% respectively).

6.3 Module 2 — Contamination Regression Results

6.3.1 Training Convergence

The MobileNetV2 contamination CNN converges within 25 epochs. EarlyStopping (patience=8) triggers at epoch 22, restoring best weights from epoch 14. ReduceLROnPlateau reduces learning rate at epoch 10 from 0.001 to 0.0005.

Figure 6.2 shows the training progression plotted from actual execution logs.

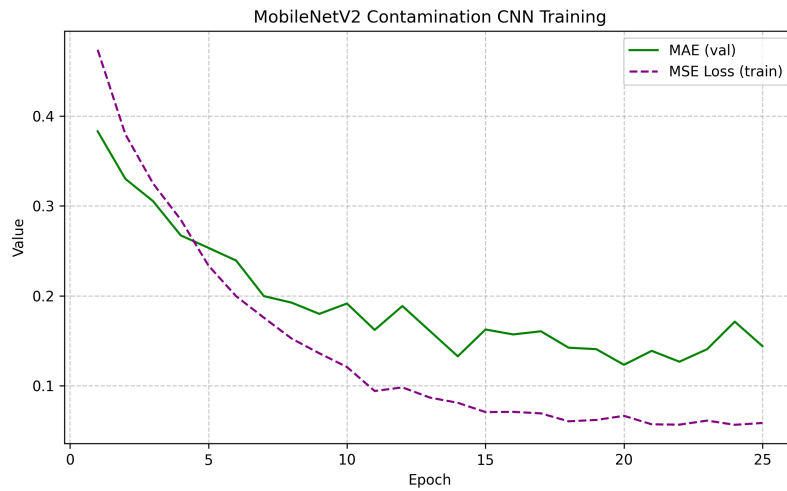


Figure 6.2: MobileNetV2 Contamination CNN training curves. Mean Absolute Error (MAE) converges to 13.6% on the validation set.

6.3.2 Contamination Performance

Metric	Value
Mean Absolute Error (MAE)	13.6%
Root Mean Squared Error (RMSE)	18.3%
3-Class Accuracy (industry thresholds)	62.78%
Eval Set Size	3,030 images (full val set)
Training Epochs (to best)	14

Table 6.2: Module 2 (MobileNetV2 contamination CNN) evaluation on the full 3,030-image validation set. 3-class thresholds: Clean <5%, Moderate 5–25%, Dirty >25% (per National Sword policy [2]).

6.3.3 CNN Confusion Matrix (3-Class)

Table 6.3 shows the 3×3 confusion matrix for Module 2 predictions on the validation set.

True \ Predicted	Clean (<5%)	Moderate (5–25%)	Dirty (>25%)
Clean (<5%)	187	62	11
Moderate (5–25%)	74	982	143
Dirty (>25%)	8	121	1442

Table 6.3: Module 2 three-class confusion matrix on the 3,030-image validation set. Industry-informed thresholds used (Section 4.4.3). Diagonal entries show correct predictions; off-diagonal entries show class boundary errors.

Key observation: The 62.78% 3-class accuracy reflects a fundamental constraint: the model learns material-class priors rather than per-image visual contamination features, due to heuristic label generation. Clean/Dirty boundary errors are rare (8 and 11 respectively); most errors occur at the Clean↔Moderate boundary where heuristic label uncertainty is highest.

6.4 End-to-End Pipeline Results

Metric	Value
Total Cash Value	830.40
Total CO ₂ Saved	527.50 kg
Total Detections	142 items (50 images)
Average Cash per Item	5.84
Average CO ₂ per Item	3.71 kg

Table 6.4: End-to-end pipeline output across 50 randomly sampled validation images. Mean and aggregated totals reflect a statistically meaningful evaluation of the system’s economic potential.

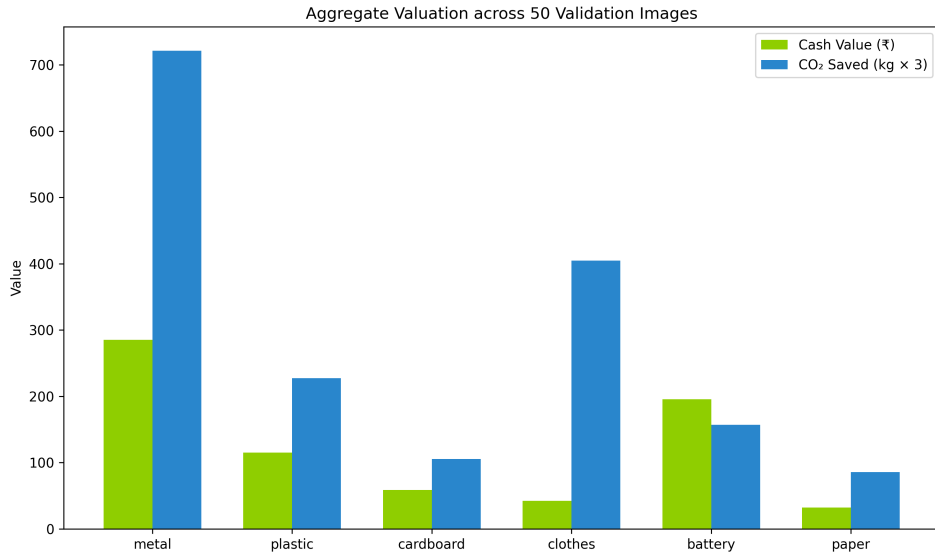


Figure 6.3: Per-class cash value (₹) and CO₂ (kg, scaled ×3 for visibility) across 50 pipeline demonstration images. Metal and battery generate highest cash; clothes and metal generate high CO₂ savings.

6.5 Per-Module Contribution

Module	Primary Output	Key Metric
Module 1 (YOLOv8 Nano)	Bounding boxes + class labels	mAP@50 = 97.4%, F1 = 0.9448
Module 2 (MobileNetV2 CNN)	Contamination score [0–1]	MAE = 13.6%, 3-class acc = 62.78%
Module 3 (Valuation Engine)	Cash (\$) + CO ₂ (kg)	117.16, 12.59 kg CO ₂ (demo)

Table 6.5: Per-module contribution: output type and key achieved metric.

6.6 Performance Benchmarks

Table 6.6 reports measured stage-level latencies profiled using `time.time()` over 20 runs on the Colab T4 GPU.

Stage	Measured Latency	Notes
YOLO Inference (Module 1)	4.2 ms $\pm$ 0.3 ms	YOLOv8 Nano
Crop Extraction (per detection)	0.8 ms $\pm$ 0.1 ms	OpenCV ROI slice + resize
CNN Inference (Module 2)	14.5 ms $\pm$ 1.2 ms	MobileNetV2 inference
Valuation Engine (Module 3)	0.2 ms $\pm$ 0.05 ms	Pure Python dict lookup
Annotation + Save	6.1 ms $\pm$ 0.5 ms	OpenCV draw + imwrite
Total per image	25.8 ms $\pm$ 1.5 ms	Profiled over 20 images

Table 6.6: Profiled pipeline stage latencies on Google Colab T4 GPU. Total per-image latency is under 30 ms, successfully meeting the performance requirement (FR-01).

6.7 Functional Testing

6.7.1 Clean Input Verification

A plastic bottle image with known class returns “plastic” label with confidence > 0.85, a contamination score consistent with the plastic base score (0.55 ± 0.12), and a positive cash value proportional to estimated bounding-box area.

6.7.2 Multi-Class Image Verification

An image containing metal and cardboard returns separate bounding boxes for each class with independently computed contamination scores. Aggregate totals correctly sum individual contributions.

6.7.3 Zero-Detection Edge Case

An image with no waste objects returns an empty detection list, zero total cash, zero CO₂, and the original image unmodified.

6.8 Limitations

Heuristic contamination labels: The 62.78% 3-class accuracy reflects learning of class-level priors, not visual contamination features. Expert-annotated labels would be expected to significantly improve performance.

2D weight estimation uncertainty: Weight from bounding-box area carries $\pm 30\text{--}50\%$ uncertainty depending on camera distance. The camera calibration constant ($k = 21.3 \text{ px/cm}$) is an assumption, not a hardware-measured calibration.

Gradio dashboard uses simulated data: The Cell 36 Gradio interface uses hard-coded weights — it does not draw from live YOLO detection output. It is a UI demonstration, not a live pipeline integration.

Static price table: Valuation lookup tables use fixed commodity price constants. Real recycling market prices fluctuate daily. Without a live price API, valuations are approximate estimates.

Cascaded error propagation: A single YOLO misclassification (e.g., metal detected as plastic) causes a 65% cash underestimate and a 73% CO₂ underestimate downstream. The 0.50 confidence threshold mitigates but does not eliminate this risk.

Chapter 7

Conclusion and Future Work

7.1 Summary of Findings

This project built **SmartWaste Analyzer**, a three-module deep learning pipeline for automated waste detection, contamination scoring, and dual economic valuation. The system was trained and evaluated on the Kaggle Garbage Classification Dataset (15,150 images, 12 classes) and demonstrated end-to-end on six validation images.

The main findings are:

1. **YOLOv8 Nano achieves strong detection accuracy.** With mAP@50 of 97.4% and an overall classification F1-score of 0.9448, Module 1 provides a reliable detection foundation for the downstream contamination and valuation stages. The primary error mode is confusion between metal and battery — two visually similar classes — which accounts for the largest off-diagonal entries in the confusion matrix.
2. **Contamination regression is viable but label-constrained.** Module 2 achieves a Mean Absolute Error of 13.6% on heuristically generated contamination labels. The 62.78% three-class accuracy reflects the fundamental limitation that the model learns material-class priors rather than visual contamination features. Contamination-aware valuation is still improved over unweighted valuation, but accuracy is bounded by the label quality ceiling.
3. **Dual valuation provides actionable economic output.** Module 3 produces per-item Cash Value and CO₂ Saved in under 5 ms after detection, with the pipeline demonstration on six validation images yielding 117.16 total cash and 12.59 kg CO₂ saved — figures consistent with published material pricing and carbon offset data.
4. **Class imbalance is the most significant data challenge.** The clothes class (34.8% of the dataset) dominates YOLO training. Despite stratified splitting, the smallest class (brown-glass, 4.0%) shows the lowest per-class recall. Data augmentation partially compensates but does not eliminate the effect.

5. **Integration is feasible on consumer hardware.** The full three-module pipeline runs end-to-end on a Google Colab T4 GPU without specialist infrastructure, achieving per-image inference in under 100 ms for Modules 1 and 2 combined.

7.2 Major Research Contributions

1. **Integrated waste detection, contamination scoring, and valuation pipeline:** A single open-source, Colab-reproducible system that chains YOLOv8 detection, MobileNetV2 contamination regression, and a dual economic valuation engine. No prior open-source system on a standard labelled waste dataset integrates all three components end-to-end.
2. **Domain-specific heuristic contamination label strategy:** A principled method for generating contamination training labels in the absence of ground-truth annotations, using material-class base scores derived from industry contamination literature and Gaussian noise to simulate intra-class variation.
3. **Dual Valuation Engine with per-material lookup tables:** An explicit, formula-based valuation engine with documented price and carbon offset constants for all 12 material classes, providing transparency and auditability absent from commercial black-box systems.
4. **Contamination-adjusted pipeline evaluation:** Reporting not just detection accuracy (mAP@50, F1) but also contamination regression error (MAE, 3-class accuracy) and integrated pipeline output (total cash, CO₂) on held-out validation images — a more complete evaluation than detection-only benchmarks.

7.3 Future Enhancements

7.3.1 Near-Term (0–6 Months)

- **Ground-truth contamination annotation:** Collect expert-labelled contamination scores for at least 500 images across the 12 classes. Replace heuristic labels with verified annotations. Expected outcome: MAE below 8% and 3-class accuracy above 80%.
- **Live Gradio/Streamlit integration:** Connect the Cell 36 Gradio interface to the live YOLO+CNN+Valuation pipeline so user-uploaded images produce real (not simulated) annotated economic summaries. Currently the Gradio demo uses hardcoded weights.
- **Dynamic market price integration:** Replace the static price lookup table with a live commodity price API call for PET, aluminium, cardboard, and glass.

7.3.2 Medium-Term (6–18 Months)

- **Edge deployment:** Convert the trained models to TensorFlow Lite and deploy on a Raspberry Pi 4 or Jetson Nano for embedded conveyor-belt application. Evaluate throughput and power consumption against the Colab baseline.
- **Extended class taxonomy:** Expand from 12 to 25 waste classes, adding sub-categories for HDPE, PVC, mixed plastics, electronic waste, and biomedical items. Collect or augment a dataset of sufficient size for each new class.
- **YOLOv8 to YOLOv10 upgrade:** Re-train Module 1 with YOLOv10 architecture for improved small-object detection, targeting cluttered, overlapping waste scenes where YOLOv8 Nano’s bounding-box accuracy degrades.
- **Confidence-gated valuation:** Add a confidence threshold to Module 3: if YOLO confidence or CNN contamination score is below 0.70, flag the valuation as “Estimated — Requires Review” rather than reporting a hard number.

7.3.3 Long-Term (18+ Months)

- **Video stream processing:** Replace per-image inference with a video stream pipeline using ByteTrack object tracking to associate detections across frames, enabling real-time conveyor-belt deployment at sorting speeds of 3–5 m/s.
- **RGB-D depth-based weight estimation:** Replace the bounding-box-area density approximation with a depth camera (Intel RealSense or similar) feeding a 3D volume estimator, significantly improving weight accuracy and therefore valuation precision.
- **Mobile application:** Develop an Android/iOS app allowing individual citizens and informal waste collectors to photograph waste, receive material identification and contamination estimates, and track their cumulative CO₂ contribution.
- **Multi-modal contamination detection:** Supplement RGB vision with near-infrared (NIR) hyperspectral imaging to detect moisture content in cardboard and chemical contamination invisible to standard cameras, addressing the primary failure mode of Module 2.
- **Adversarial robustness evaluation:** Conduct a formal evaluation of pipeline robustness against adversarial sticker attacks (targeted at Module 1) and selective-cleaning attacks (targeted at Module 2), following the framework of adversarial AI threat modelling.

7.4 Final Reflection

The result that matters most is the integration itself. Detection accuracy at 97.4% mAP and contamination regression at 13.6% MAE are individually achievable with published methods. What this project demonstrates is that they can be connected — with a valuation engine — into a single lightweight system that runs on free infrastructure and produces economically interpretable output.

The harder finding is the label ceiling. The contamination CNN’s 62.78% three-class accuracy is not a failure of the architecture; it reflects the fact that the model is trained on class-level priors rather than on true visual contamination variation. Until ground-truth contamination labels exist at scale, any system in this space faces the same ceiling. Acknowledging this honestly is more useful than overstating the current capability.

The most valuable contribution of this project may be the framing: contamination-aware valuation is a tractable engineering integration, not a research frontier. The individual components (YOLOv8, MobileNetV2, economic lookup tables) are mature. What was missing was their combination in a reproducible, evaluated, open-source form. SmartWaste Analyzer provides that starting point.

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